

## Performance Testing

|               |                                                          |
|---------------|----------------------------------------------------------|
| Date          | 02 October 2026                                          |
| Team ID       | SWTID-2026-3464                                          |
| Project Name  | FitBuddy – AI Fitness Plan Generator using Gemini Models |
| Maximum Marks | 5 Marks                                                  |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field               | Details                        |
|---------------------|--------------------------------|
| Testing Tool Used   | Postman                        |
| Type of Testing     | Load / API Performance Testing |
| Target Module / API | Workout Plan Generation API    |
| Test Environment    | Local                          |
| Test Date           | 02 October 2026                |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description                   | No. of Virtual Users | Duration (sec) | Expected Outcome                                               |
|------|-----------------------------------------------|----------------------|----------------|----------------------------------------------------------------|
| 1    | Single workout-plan generation request        | 1                    | 14.17          | Workout plan generated successfully                            |
| 2    | Multiple consecutive plan-generation requests | Not tested           | Not tested     | Requests should be processed successfully without errors       |
| 3    | Feedback-based plan update request            | Not tested           | Not tested     | Existing plan should be updated successfully based on feedback |
| 4    | Database retrieval/request                    | Not tested           | Not tested     | Stored workout information should be retrieved successfully    |

**Step 3: Performance Test Results**

| S.No | Metric               | Target Value | Actual Value  | Status (Pass / Fail) | Remarks                          |
|------|----------------------|--------------|---------------|----------------------|----------------------------------|
| 1    | Response Time (Avg)  | < 15 seconds | 14.17 seconds | PASS                 | Response generated successfully  |
| 2    | Response Time (Max)  | < 20 seconds | 14.17 seconds | PASS                 | Within defined maximum limit     |
| 3    | Throughput (Req/sec) | —            | 0.071 req/sec | PASS                 | Based on 1 successful request    |
| 4    | Error Rate           | < 1%         | 0%            | PASS                 | 1 successful request, 0 failures |
| 5    | CPU Utilization      | < 80%        | Not measured  | —                    | Not captured during this test    |
| 6    | Memory Utilization   | < 80%        | Not measured  | —                    | Not captured during this test    |

**Step 4: Observations & Analysis**

**Key Findings**

The FitBuddy application successfully processed the tested API requests and generated workout-plan responses. The observed response time and error rate were analyzed under the tested workload. Performance may vary depending on Gemini API response time and the test environment.

**Bottlenecks Identified**

The main potential bottleneck is the dependency on the Gemini API, since AI response generation can introduce variable response time.

**Optimization Steps Taken**

The application sends the required fitness information and feedback to the backend for processing and uses SQLite for lightweight data storage. Input validation is performed before processing requests.

## Step 5: Screenshots / Evidence

